Business Rules for BazaarX Ver. 1

Note: Temporary BRD for Email and SMS Automation; Can still be updated

Module 1 - Email & SMS Automation

The Email & SMS Automation Module governs the creation, triggering, delivery, monitoring, and compliance of all system-generated communications within the e-commerce platform.

This module ensures:

a. Timely transactional notifications
b. Secure delivery of authentication messages
c. Controlled execution of marketing campaigns
d. Regulatory compliance with data privacy and anti-spam laws
e. Proper audit logging and traceability
f. Prevention of communication abuse or system misuse

The module applies to both customer-facing and system-generated notifications across Email and SMS channels.

Controller Assignment

This section defines the designated role or authority responsible for governing, executing, monitoring, and enforcing each automation rule within the Email & SMS Automation module.

Section	Section Title	Primary Controller	Secondary Controller	Rationale
A	Rule Classification & Communication Governance	Platform System (Core Logic)	Super Admin (Policy Configuration)	Classification, consent enforcement, quiet hours, and frequency limits must be system-enforced to prevent compliance risk. Admin may configure thresholds but cannot bypass logic.
B	Transactional Automation Rules	Platform System	None (Non-Overridable)	Order, payment, shipment, and refund notifications are operationally mandatory and must not be manually controlled or disabled by sellers.

C	OTP & Security Rules	Platform System	Super Admin (Security Policy Settings)	OTP expiry and rate limiting are security controls and must be system-managed. Admin may configure expiry duration and retry limits but cannot manually trigger or bypass validation.
D	Template Governance & Content Control	Super Admin (Marketing/Admin Team)	Platform System (Approval Enforcement)	Templates require human governance for branding and legal compliance. The system enforces approval workflow and prevents unapproved usage.
E	Delivery & Failure Handling	Platform System	DevOps / Technical Admin (Monitoring)	Retry logic, bounce suppression, and queuing must be automated. Technical teams monitor delivery infrastructure but do not manually trigger retries.
F	Logging, Audit & Compliance	Platform System	Compliance/Admin (Review Access)	Logging must be automatically enforced. Admin and Compliance teams may review logs but cannot alter historical records.

A. Rule Classification

This section defines how communications are categorized, controlled, and governed before being sent. It establishes consent enforcement, marketing limitations, and message classification standards.

Rule ID	Rule Title	Rule Statement	Trigger/Event	Conditions	System Enforcement	Applicable Channel

BR-EMA-001	Communication Categorization	All outgoing communications shall be classified as Transactional, Security, or Marketing prior to dispatch.	Message creation	Category field is required	System shall block dispatch if category is null or invalid.	Email
BR-EMA-002	Mandatory Transactional Delivery	Transactional communications shall be delivered regardless of marketing consent status.	Transactional event	Event successfully recorded	System shall bypass marketing suppression logic.	Email
BR-EMA-003	Marketing Consent Requirement	Marketing communications shall only be sent to users with recorded explicit consent.	Campaign execution	User consent = Active	System shall suppress communication if consent is inactive or revoked.	Email
BR-EMA-004	Unsubscribe Enforcement	Users shall be able to revoke marketing consent at any time.	Unsubscribe action	Valid user identity	System shall immediately update consent status and suppress future marketing communications.	Email
BR-EMA-005	Quiet Hours Restriction	Marketing communications shall not be delivered between 21:00 and 08:00 based	Campaign dispatch	Marketing classification	System shall defer delivery until permissible hours.	Email

		on user timezone.				
BR-EMA-006	Frequency Limitation	Marketing communications shall not exceed defined weekly thresholds.	Campaign scheduling	Email ≤ 3/week; SMS ≤ 2/week	System shall block dispatch if threshold exceeded.	Email

B. Transactional Automation Rules

This section defines mandatory automated notifications triggered by order lifecycle events and financial transactions. These notifications are critical for order transparency, payment confirmation, and operational trust.

Rule ID	Rule Title	Rule Statement	Trigger/Event	Conditions	System Enforcement	Applicable Channel
BR-EMA-007	Order Confirmation Notification	Order confirmation notification shall be sent upon successful order creation.	Order created	Payment status = Pending or Paid	System shall send notification within 60 seconds.	Email
BR-EMA-008	Payment Confirmation Notification	Payment confirmation shall be sent upon successful payment verification.	Payment verified	Payment status = Completed	System shall send notification immediately.	Email
BR-EMA-009	Shipment Notification	Shipment notification shall be sent when order status	Status update	Tracking number available	System shall include tracking details in communication.	Email

		changes to Shipped.				
BR-EMA-010	Delivery Notification	Delivery confirmation shall be sent upon confirmed delivery.	Status update	Delivery confirmed	System shall notify customer immediately.	Email, SMS (optional)
BR-EMA-011	Refund Notification	Refund confirmation shall be sent once refund is processed.	Refund processed	Refund status = Completed	System shall send notification without delay.	Email, SMS
BR-EMA-012	Security Notification	Security alerts shall be sent immediately upon detection of security events.	Password reset, suspicious login	Security event validated	System shall prioritize dispatch without delay.	Email, SMS

C. OTP & Security Rules

This section governs the issuance, validity, and protection of One-Time Passwords (OTP) and security-related communications to ensure user authentication integrity and fraud prevention.

Rule ID	Rule Title	Rule Statement	Trigger/Event	Conditions	System Enforcement	Applicable Channel
BR-EMA-013	OTP Validity	One-Time Password (OTP) shall expire within five (5) minutes from generation.	OTP generated	Valid phone/email	System shall invalidate OTP after expiration time.	SMS, Email

BR-EMA-014	OTP Retry Limitation	OTP requests shall be limited to prevent abuse.	OTP request	Maximum 5 attempts per hour	System shall temporarily block further requests upon limit breach.	SMS, Email

D. Template Governance & Content Control

This section establishes content governance standards, template approval requirements, version control, personalization validation, and legal compliance requirements for outbound communications.

Rule ID	Rule Title	Rule Statement	Trigger/Event	Conditions	System Enforcement	Applicable Channel
BR-EMA-015	Template Approval Requirement	All templates must be approved prior to activation.	Template creation/edit	Status = Approved by Admin	System shall prevent use of draft or unapproved templates.	Email, SMS
BR-EMA-016	Template Version Control	Template modifications shall create a new version record.	Template update	Existing live template	System shall preserve historical versions.	Email, SMS
BR-EMA-017	Personalization Validation	Dynamic variables must be validated before message dispatch.	Message generation	Required fields populated	System shall block dispatch if required variables are missing.	Email
BR-EMA-018	Legal Footer Inclusion	Marketing communications shall include unsubscribe mechanism and legal disclosure.	Marketing send	Message classified as Marketing	System shall append mandatory footer.	Email

E. Delivery & Failure Handling

This section defines retry mechanisms, bounce suppression rules, queue management during downtime, and delivery assurance protocols to maintain communication reliability.

Rule ID	Rule Title	Rule Statement	Trigger/Event	Conditions	System Enforcement	Applicable Channel
BR-EMA-019	Email Retry Logic	Failed email deliveries shall be retried up to three (3) times.	Delivery failure	Provider response = Failed	System shall attempt retries before marking failure.	Email
BR-EMA-020	SMS Retry Logic	Failed SMS deliveries shall be retried once.	Delivery failure	Provider response = Failed	System shall attempt one retry only.	SMS
BR-EMA-021	Bounce Suppression	Contacts with three (3) consecutive hard bounces shall be marked invalid.	Bounce detected	Bounce threshold reached	System shall suppress future communications.	Email
BR-EMA-022	Queue During Downtime	Messages generated during service downtime shall be queued.	Automation service unavailable	System outage detected	System shall dispatch queued messages upon recovery.	Email, SMS

F. Logging, Audit & Compliance

This section ensures transparency, traceability, performance monitoring, regulatory alignment, and record retention for all automated communications.

Rule ID	Rule Title	Rule Statement	Trigger/Event	Conditions	System Enforcement	Applicable Channel

BR-EMA-023	Communication Logging	All outgoing communications shall be logged.	Message sent	Valid send attempt	System shall log recipient, timestamp, template ID, delivery status.	Email, SMS
BR-EMA-024	Log Retention	Communication logs shall be retained for minimum twelve (12) months.	Log storage	Log created	System shall enforce retention policy.	Email, SMS
BR-EMA-025	Data Privacy Compliance	Communications shall comply with applicable data protection and anti-spam regulations.	Any dispatch	User data processed	System shall enforce consent tracking and data security standards.	Email, SMS
BR-EMA-026	Performance Monitoring	Campaign performance metrics shall be recorded and reportable.	Campaign execution	Valid campaign ID	System shall track open rate, click rate, unsubscribe rate, and delivery rate.	Email

G. Contact Validation

This section ensures that all contact information is validated prior to message dispatch to prevent unnecessary delivery failures and protect sender reputation.

Rule ID	Rule Title	Rule Statement	Trigger/Event	Conditions	System Enforcement	Applicable Channel
BR-EMA-027	Contact Format Validation	All email addresses and phone numbers shall be validated before	Message creation or send attempt	Contact information present	System shall validate email format and phone number structure before sending.	Email, SMS

		message dispatch.				
BR-EMA-028	Invalid Contact Blocking	Messages shall not be dispatched to contacts marked as invalid.	Send attempt	Contact status = Invalid	System shall block dispatch and log suppression reason.	Email, SMS
BR-EMA-029	Country Code Standardization	SMS numbers shall follow international format (E.164 standard).	Phone number entry	SMS channel selected	System shall normalize and validate phone number before send.	SMS

H. Bounce Suppression Enhancement (Hard vs Soft)

This section differentiates between permanent and temporary delivery failures to protect sender reputation while allowing retry for recoverable errors.

Rule ID	Rule Title	Rule Statement	Trigger/Event	Conditions	System Enforcement	Applicable Channel
BR-EMA-030	Hard Bounce Suppression	Contacts with a hard bounce shall be immediately marked as invalid.	Delivery failure	Failure type = Hard Bounce	System shall suppress future email sends.	Email
BR-EMA-031	Soft Bounce Retry Policy	Soft bounces shall trigger retry attempts before suppression.	Delivery failure	Failure type = Soft Bounce	System shall retry per retry logic before suppression.	Email

BR-EMA-032	Soft-to-Hard Conversion	Three (3) consecutive soft bounces shall convert to hard suppression status.	Bounce tracking	Soft bounce threshold reached	System shall mark contact as invalid.	Email

Recommended Industry Standard:
Immediate suppression for hard bounces; 2–3 retry attempts for soft bounces.

I. Consent Timestamp Logging
This section ensures that all marketing consent actions are traceable for compliance and audit purposes.

Rule ID	Rule Title	Rule Statement	Trigger/Event	Conditions	System Enforcement	Applicable Channel
BR-EMA-033	Consent Timestamp Logging	The system shall record the timestamp of all opt-in and opt-out actions.	Consent update	User action recorded	System shall store date/time and consent status.	Email, SMS
BR-EMA-034	Consent Source Tracking	The system shall record the source of consent acquisition.	Consent capture	Signup, campaign, manual entry	System shall store source metadata.	Email, SMS
BR-EMA-035	Immutable Consent Records	Consent records shall not be editable once stored.	Consent update	Record created	System shall preserve historical consent logs.	Email, SMS

J. Campaign Approval Workflow
This section defines approval requirements before campaign activation to ensure brand consistency and compliance.

Rule ID	Rule Title	Rule Statement	Trigger/Event	Conditions	System Enforcement	Applicable Channel

BR-EMA-036	Campaign Approval Requirement	Marketing campaigns must be approved before activation.	Campaign creation	Status = Pending Approval	System shall prevent launch without approval.	Email, SMS
BR-EMA-037	Scheduled Campaign Lock	Approved campaigns scheduled for dispatch shall be locked from editing.	Scheduled campaign	Dispatch time set	System shall prevent edits unless reverted to draft.	Email, SMS
BR-EMA-038	Emergency Campaign Suspension	Admin shall have authority to suspend active campaigns.	Manual intervention	Admin role validated	System shall immediately halt campaign dispatch.	Email, SMS

K. Blacklist / Suppression List Governance

This section governs permanent suppression of contacts due to fraud, abuse, or regulatory requirements.

Rule ID	Rule Title	Rule Statement	Trigger/Event	Conditions	System Enforcement	Applicable Channel
BR-EMA-039	Global Suppression List	The system shall maintain a global suppression list.	Suppression event	Contact flagged	System shall block all marketing communications.	Email, SMS
BR-EMA-040	Manual Blacklisting	Authorized Admin may manually blacklist a contact.	Admin action	Valid justification	System shall suppress future communications.	Email, SMS

BR-EMA-041	Suppression Override Restriction	Blacklisted contacts shall not receive marketing messages even if consent is active.	Send attempt	Contact status = Blacklisted	System shall block dispatch.	Email, SMS

L. SLA & Delivery Time Standards

This section defines acceptable delivery performance benchmarks to ensure operational reliability.

Rule ID	Rule Title	Rule Statement	Trigger/Event	Conditions	System Enforcement	Applicable Channel
BR-EMA-042	Transactional SLA	Transactional notifications shall be dispatched within sixty (60) seconds of event confirmation.	Event trigger	System operational	System shall prioritize transactional queue.	Email, SMS
BR-EMA-043	Marketing Delivery Window	Marketing messages shall be delivered within scheduled timeframe ±5 minutes.	Campaign dispatch	Scheduled time reached	System shall monitor delivery delay.	Email, SMS
BR-EMA-044	Queue Threshold Alert	If pending message queue exceeds defined threshold, alert shall be triggered.	Queue monitoring	Threshold breached	System shall notify DevOps/Admin.	Email, SMS

M. Escalation & Incident Handling
This section defines escalation protocols when delivery infrastructure or automation services fail.

Rule ID	Rule Title	Rule Statement	Trigger/Event	Conditions	System Enforcement	Applicable Channel
BR-EMA-045	Provider Failure Escalation	Extended messaging provider failure shall trigger incident escalation.	Delivery failure spike	Failure > defined threshold	System shall notify technical team.	Email, SMS
BR-EMA-046	Domain Reputation Monitoring	Significant drop in deliverability rate shall trigger review.	Performance monitoring	Bounce rate > threshold	System shall flag compliance review.	Email
BR-EMA-047	Incident Logging	All automation-related incidents shall be logged.	System error	Error recorded	System shall log timestamp and resolution status.	Email, SMS

N. Role-Based Access Control (RBAC)
This section defines access limitations and permission controls to prevent unauthorized manipulation of automation settings.

Rule ID	Rule Title	Rule Statement	Trigger/Event	Conditions	System Enforcement	Applicable Channel
BR-EMA-048	Role Restriction Enforcement	Automation controls shall be accessible only to authorized roles.	Access attempt	Role validation	System shall restrict access by role permissions.	Email, SMS
BR-EMA-049	Log Immutability	Communication logs shall not be editable by any role.	Log access	Any role	System shall enforce read-only access.	Email, SMS

BR-EMA-050	Seller Limitation	Sellers shall not modify transactional, security, or consent enforcement rules.	Seller access attempt	Seller role	System shall block configuration access.	Email, SMS